Closing Three Open Anatomical Questions in the Figure-Ground Circuit

Stage 6 – Open Questions

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Abstract

We resolve three open anatomical questions left by the figure-ground circuit report (VCH→T4/T5→LLPC1→Nod1→steering) end to end from connectome data, each yielding a neutral, falsifiable verdict. single-animal closure: CONFIRMED; bilateral generalization: CONFIRMED, UNVERIFIABLE; escape-route separability: CONFIRMED. Across all claims the tally is CONFIRMED=19, UNVERIFIABLE=1. Every number is re-derived from the FlyWire FAFB v783 live CAVE synapse table (`synapses_nt_v1`, no cleft threshold) and the male whole-CNS (MCNS) v1.0 offline bulk feathers; negatives are reported faithfully.

1 Q1 – Single-Animal Closure of the LLPC1→Nod1→DNp26 Readout

Question

Does the figure-ground readout chain close inside a SINGLE animal? The published circuit is split across two specimens of different sex (female FlyWire FAFB for the optic-lobe/DN side, male MCNS for the motor side). We test whether LLPC1, Nod1 and DNp26 exist in MCNS and whether the LLPC1→Nod1→DNp26 wiring reproduces within MCNS alone, and report the fraction of the chain recovered in one animal.

Methods

Both connectome layers are queried offline from the MCNS v1.0 public bulk feathers (`body-annotations.feather`, `connectome-weights.feather`; synapse counts as edge weights). Cell identity is resolved through the `type` column; the LLPC1→Nod1 and Nod1→DNp26 edges are summed over all bodies of each type. The within-MCNS motif is compared row-for-row against the paper's FlyWire numbers (Nod1 the dominant excitatory readout; Nod1→DNp26 the strongest convergent steering target).

Results

Quantity	Value
question	Q1__single__animal__closure
existence	LLPC1=285, Nod1=4, DNp26=2, DNa04=2, DNbe001=2
wiring	llpc1__to__nod1__syn=10871, llpc1__contacting__nod1=272, nod1__to__dnp26__syn=504, nod1__contactin...
nod1__readout__rank	nod1__rank=2, nod1__syn=10871, n__target__types=673, top5__targets=[{'type': 'PLP249', 'weight...
chain__reproduced	1
verdict	CONFIRMED
track	mcns

Verdict

Overall: **CONFIRMED**

ID	Description	Report	Computed	V
`Q1.llpc1_present`	LLPC1 present in MCNS (>=1 body)	>=1 body (~285)	True	C
`Q1.nod1_present`	Nod1 present in MCNS (>=1 body)	>=1 body (~4)	True	C
`Q1.dnp26_present`	DNp26 present in MCNS (>=1 body)	>=1 body (~2)	True	C
`Q1.llpc1_to_nod1_edge`	LLPC1 -> Nod1 synaptic edge exists in MCNS (syn>0)	syn>0	True	C
`Q1.nod1_to_dnp26_edge`	Nod1 -> DNp26 synaptic edge exists in MCNS (syn>0)	syn>0	True	C
`Q1.nod1_dominant_readout`	Nod1 among LLPC1's top excitatory targets in MCNS	top excitatory readout	True	C
`Q1.chain_reproduced`	Fraction of LLPC1->Nod1->DNp26 chain reproduced within MCNS alone	1.0 (full chain, single animal)	1	C

2 Q2 – Bilateral / Four-Direction Generalization of the Sheet

Question

Do the headline numbers GENERALIZE across hemisphere and direction, or are they specific to the shown exemplar (right LLPC1 sheet, front-to-back layer-a T4a)? We derive the LEFT LLPC1 sheet (and, where feasible, the other directional T4/T5 channels) and test, per hemisphere, whether (a) VCH presynaptically gates the driving terminals, (b) pooling is spatially local/retinotopic, and (c) Nod1 dominates the readout.

Methods

The brain side is queried live from FlyWire FAFB v783 via CAVE (`synapses_nt_v1`, no cleft threshold, reproducing the paper counts exactly). Positions are in nanometres. The LEFT sheet uses the RIGHT-soma VCH/DCH centrifugal cells (centrifugal cells cross, so the right VCH gates the left sheet). Retinotopic locality is assessed on synapse positions; each headline number is computed per hemisphere and compared to its right-sheet analogue.

Results

Quantity	Value
question	Q2__bilateral__generalization
right	sheet__side=right, gating__soma__side=left, gating__present=True, n__vch=1, n__dch=1, vch__in__sy...
left	sheet__side=left, gating__soma__side=right, gating__present=True, n__vch=1, n__dch=1, vch__in__sy...
track	live

Verdict

Overall: **CONFIRMED** **UNVERIFIABLE**

ID	Description	Report	Computed	Verdict
`Q2.right.reciprocal`	RIGHT sheet reciprocal T4/T5 fraction (control)	89.2	89.2	**CONFIRM**
`Q2.right.vch_gating`	RIGHT sheet: VCH presynaptically gates driving terminals (control)	present	True	**CONFIRM**
`Q2.right.nod1_dominant`	RIGHT sheet: Nod1 dominant excitatory readout (control)	dominant	True	**CONFIRM**
`Q2.left.vch_gating`	LEFT sheet: VCH (right-soma) presynaptically gates driving terminals	present	True	**CONFIRM**
`Q2.left.nod1_dominant`	LEFT sheet: Nod1 dominates the readout (rank parallels right)	dominant	True	**CONFIRM**
`Q2.left.pooling_local`	LEFT sheet: pooling is spatially local / retinotopic	local	–	**UNVERIF**
`Q2.left.reciprocal_parallel`	LEFT reciprocal T4/T5 fraction parallels RIGHT (within tolerance)	~89.2% (right)	86.3	**CONFIRM**

3 Q3 – Escape-Route Census to Muscle and Its Separability

Question

Is the PLP/PVLP→LPLC2/LC4 escape arm ANATOMICALLY SEPARABLE from the LLPC1 course-control / wing-steering output? We enumerate the full descending-neuron census downstream of LPLC2/LC4 and the broadcast cells (giant-fibre DNp01 plus DNp02/03/04/06), trace them to the muscle level in MCNS with the same DN→motor-neuron→muscle method as the Nod1 arm, and test whether the two outputs use distinct DNs and distinct muscles rather than converging.

Methods

DN identity and the broadcast/escape cluster are resolved in FlyWire FAFB v783 (live CAVE); the DN→motor-neuron→muscle trace is read offline from the MCNS v1.0 feathers using the same tracer as the Nod1 arm (motor neurons annotated by innervated muscle and `somaSide`; wing-steering = subclass `wm` excluding DLM/DVM power muscles). Separability is the overlap of the escape DN/muscle sets with the LLPC1→Nod1→DNp26 sets.

Results

Quantity	Value
question	Q3_escape_route_census
flywire	n_lplc2=210, n_lc4=104, n_dn_types=28, command_dn_syn={'DNp01': 3875, 'DNp03': 741, 'DNp0...
mcns	escape_dns_in_mcns=['DNa07', 'DNae007', 'DNb05', 'DNc02', 'DNg40', 'DNge054', 'DNp01', 'D...
separability	dn_overlap=[], dn_separable=True, muscle_overlap=[], muscle_separable=True, separable=True
track	live

Verdict

Overall: **CONFIRMED**

ID	Description	Report	Co
`Q3.census_nonempty`	Escape-route DN census downstream of LPLC2/LC4 is non-empty	>=1 DN	
`Q3.looming_cluster`	Looming command cluster (DNp01/03/04/06) downstream of LPLC2/LC4	present (giant fibre DNp01 + DNp03/04/06)	

ID	Description	Report	Co
Q3.reaches_jump_muscle	Escape route reaches jump/TTM (tergotrochanter) muscle in MCNS	reaches jump/TTM	
Q3.dn_separable	Escape DN-set disjoint from steering (LLPC1) DN-set	overlap=0 (separable)	
Q3.muscle_separable	Escape muscle-targets disjoint from wing-steering muscles	overlap=0 (separable)	
Q3.escape_separable	Escape output anatomically separable from LLPC1 steering output	separable (distinct DNs + muscles)	

4 Verdict Tally

Verdict	Count
CONFIRMED	19
UNVERIFIABLE	1
Total	**20**

5 Provenance & Reproducibility

All materializations and the run environment:

Item	Value
FlyWire FAFB	{'materialization': 783}
Male CNS (MCNS)	{'materialization': 'v1.0'}
Seeds	LLPC1, Nod1, DNp26, LPLC2, LC4, DNp01, DNp03, DNp04, DNp06
Python env	flyconn_cave conda env
PYTHONPATH	src
FLYCONN_DATA_ROOT	/orcd/data/tpoggio/001/mabdel03/connectome_data

Exact command:

```
python -m flyconn.openq run && python -m flyconn.openq report
```

Token-free re-run. This PDF is regenerated from `openq_results.json` and `openq_raw.json` alone (`python -m flyconn.openq report`); the report step reads neither the live CAVE API nor the MCNS feathers, so a reviewer with only the two JSON files and a TeX engine reproduces the identical document with no FlyWire token and no network.